1. Cloud Computing – Technical Definition

Cloud computing is the **on-demand delivery of compute, storage, networking, identity, security, and platform services** over a provider-managed infrastructure, accessed via **APIs, portals, or CLI**, and billed using a **pay-as-you-consume** model.

Key Technical Points

- Resources are **virtualized** (Hyper-V in Azure)
- Provisioned using **software-defined infrastructure**
- Access controlled using **IAM (Identity & Access Management)**
- Fully **API-driven** (ARM, REST, PowerShell, CLI)

Features of Cloud Computing

High Availability (HA)

- Achieved via **redundancy**
- Azure uses:
  - Availability Zones
  - Fault domains
  - Update domains

Scalability

- **Vertical**: Increase VM size (CPU/RAM)
- **Horizontal**: Add/remove instances

Elasticity

- Automatic scaling based on:
  - CPU
  - Memory
  - Queue length
  - Schedules

Disaster Recovery

- Region-level failures
- Geo-replication
- Backup vaults

Fault Tolerance

- Service continues despite:
  - Disk failure
  - Host failure
  - Rack failure

Types of Cloud (Deployment Models)

Public Cloud

- Shared physical infra
- Logical isolation via tenants
- Azure uses **multi-tenant architecture**

Examples:

- Azure VM
- Azure SQL
- Azure Storage

Private Cloud

- Dedicated infra
- Full control
- Uses virtualization (VMware / Hyper-V)

Azure Equivalent:

Azure Stack / Azure Stack HCI

Hybrid Cloud (Most Used in Enterprises)

- On-prem + Azure connected
- Integrated using:
  - VPN
  - ExpressRoute
  - Azure Arc

📌 **AZ-900 Exam Favorite**

Multi-Cloud

- Multiple cloud vendors
- Used for:
  - Vendor lock-in avoidance
  - Compliance
  - DR

5. Cloud Service Models (CRITICAL FOR AZ-104)

IaaS – Infrastructure as a Service

What You Get

- Virtual machine
- Virtual network
- Virtual disks

What YOU Manage

- OS patching
- Firewall (inside VM)
- Applications
- Data
- Antivirus

Azure Examples

- Azure Virtual Machines
- Azure VNET
- Azure Load Balancer

📌 **Equivalent to owning a server without hardware**

PaaS – Platform as a Service

What Azure Manages

- OS
- Runtime
- Scaling
- Load balancing
- Patching

What YOU Manage

- Application code
- Data
- Identity access

Azure Examples

- Azure App Service
- Azure SQL Database
- Azure Functions

📌 **No RDP / SSH access**

SaaS – Software as a Service

Azure Manages EVERYTHING

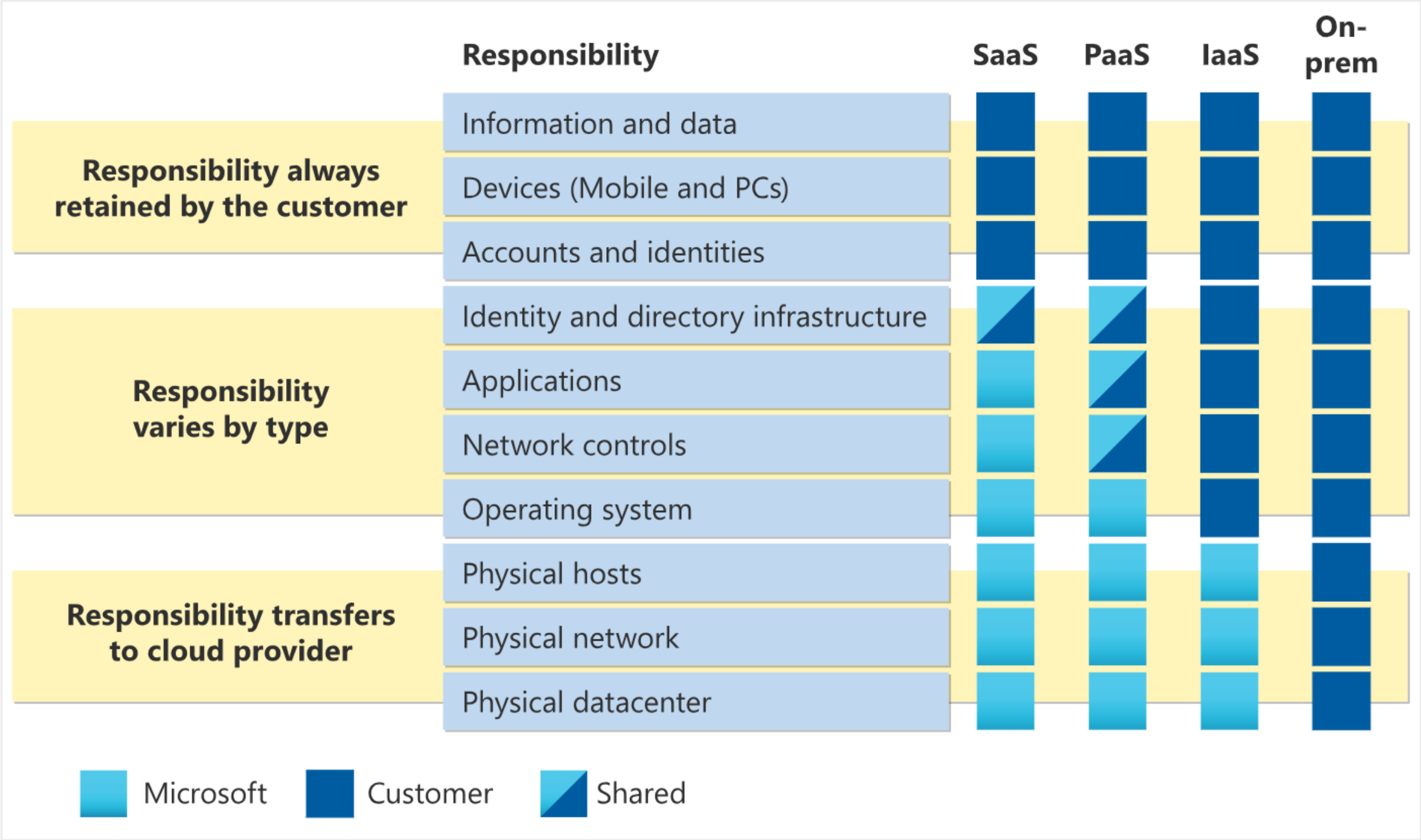
- Infrastructure
- Platform
- Application
- Security patches

You Manage

- Users
- Data access

Examples

- Microsoft 365
- Azure DevOps
- Outlook


Responsibility
SaaS
PaaS
IaaS
On-prem
Responsibility always retained by the customer
Information and data
Devices (Mobile and PCs)
Accounts and identities
Responsibility varies by type
Identity and directory infrastructure
Applications
Network controls
Operating system
Responsibility transfers to cloud provider
Physical hosts
Physical network
Physical datacenter
Microsoft
Customer
Shared

◆ What does Microsoft Azure do?

Azure provides **on-demand cloud services**, meaning you pay only for what you use and scale resources anytime.

It helps you:

- Host **virtual machines (servers)**
- Store **data securely**
- Run **applications & websites**
- Build **AI, ML, and data analytics solutions**
- Manage **networks, security, and identity**

Azure Global Infrastructure (Physical Layer)

Regions

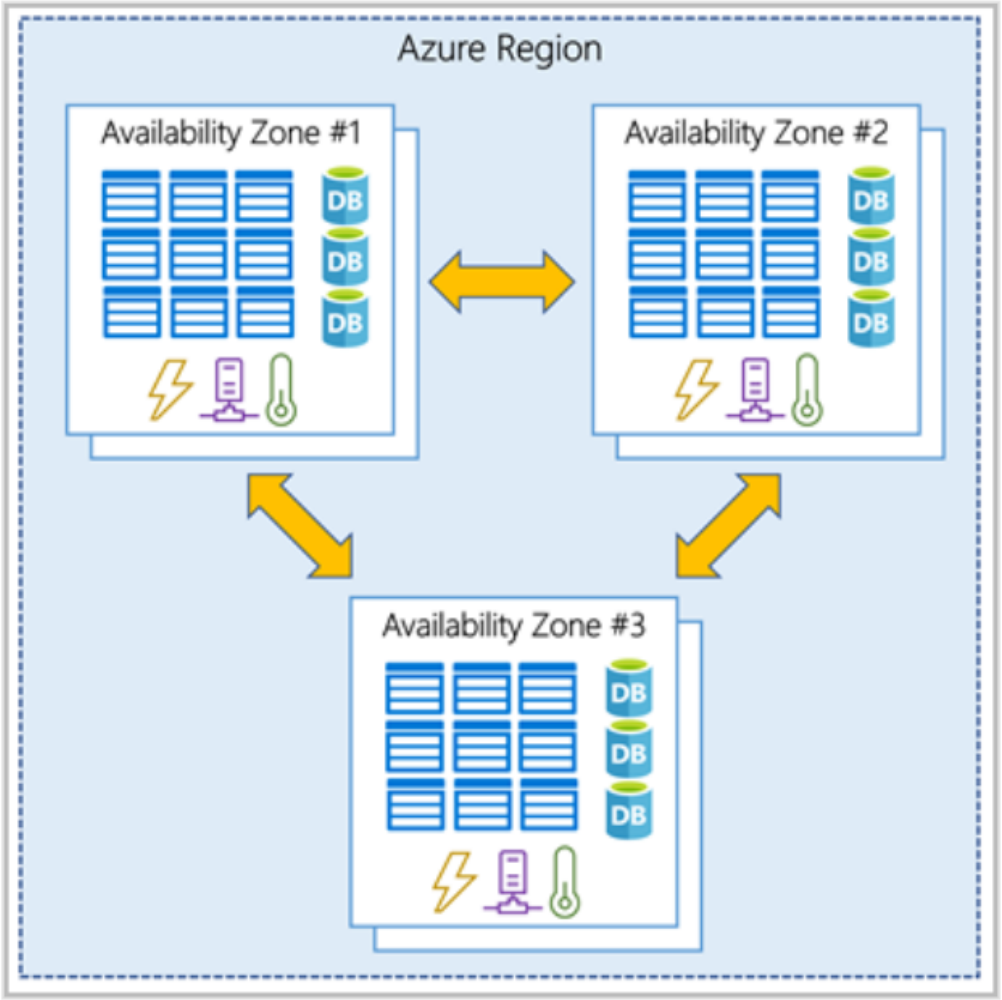
A region is a geographical area on the planet that contains at least one, but potentially multiple datacenters that are nearby and networked together with a low-latency network. Azure intelligently assigns and controls the resources within each region to ensure workloads are appropriately balanced.

When you deploy a resource in Azure, you'll often need to choose the region where you want your resource deployed.

Norway West
Norway East
West Europe
Germany West Central
UK West
Germany North
North Europe
Germany Northeast
UK South
Germany Central
France Central
Switzerland North
France South
Switzerland West
West Central US
West US 2
West US
US Gov Arizona
US Gov Texas
South Central US
US Gov Iowa,
Central US
Canada East
Canada Central
North Central US
US DoD East
East US,
East US 2,
US Gov Virginia
US DoD Central
China North,
China North 2
Korea Central
Korea South
Japan East
Japan West
China East,
China East 2
East Asia
UAE North
UAE Central
West India
Central India
South India
Southeast Asia
Brazil South
South Africa North
South Africa West
Australia East
Australia Southeast
Australia Central,
Australia Central 2
Available region
Announced region
Availability Zone(s) present

Availability Zones

Availability zones are physically separate datacenters within an Azure region. Each availability zone is made up of one or more datacenters equipped with independent power, cooling, and networking. An availability zone is set up to be an isolation boundary. If one zone goes down, the other continues working. Availability zones are connected through high-speed, private fiber-optic networks.



Azure Datacenter

- A **physical building**
- Contains:
  - Compute racks
  - Storage clusters
  - Networking equipment
- Not directly visible to customers

📌 Customers **never choose a datacenter**, only **regions**

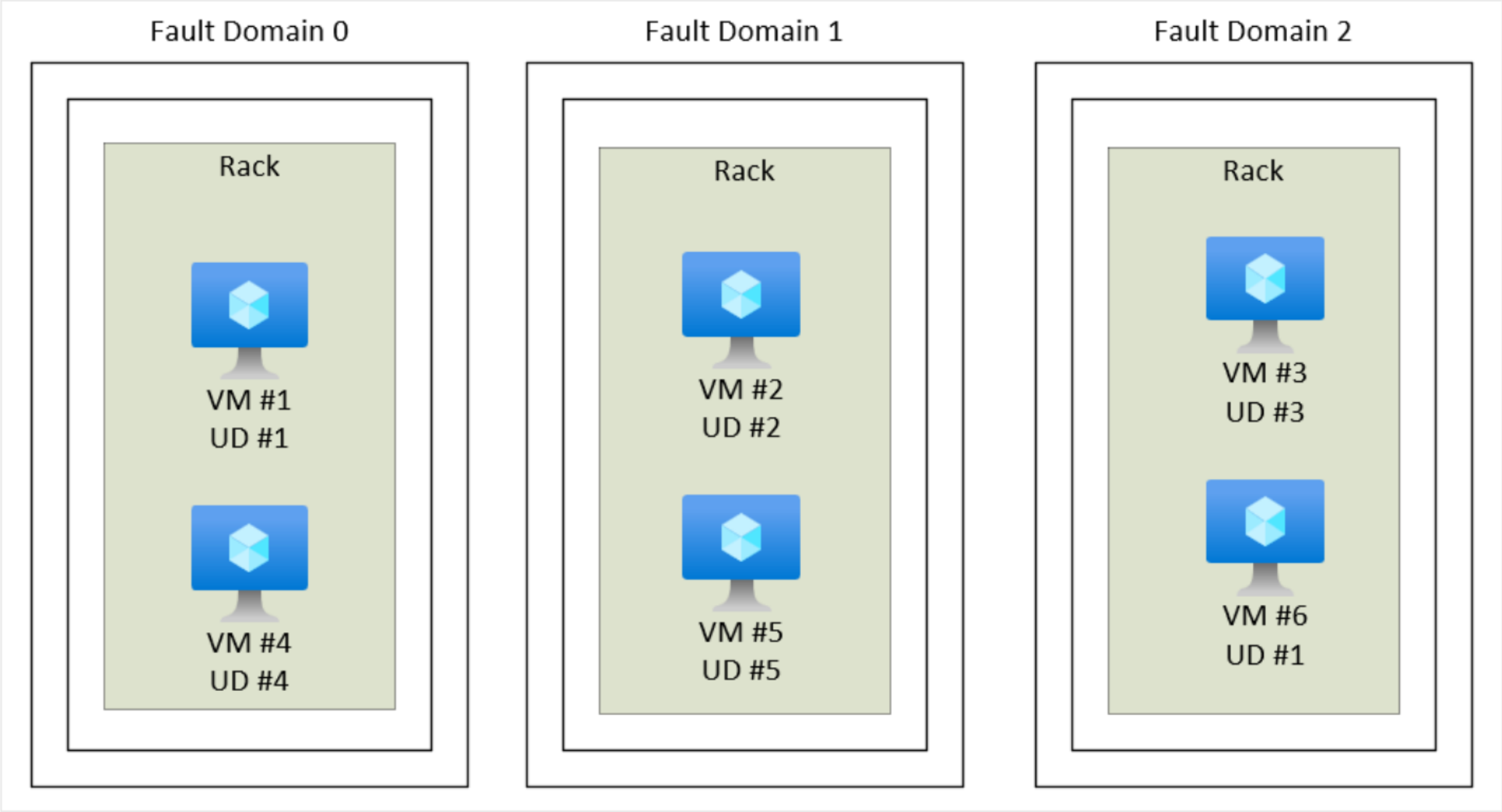
4. Availability Sets (Logical HA)

Purpose

High availability **within a single datacenter**.

Components

- **Fault Domains**
  - Protect against **hardware/rack failure**
- **Update Domains**
  - Protect against **planned maintenance**


Fault Domain 0
Rack
VM #1
UD #1
VM #4
UD #4
Fault Domain 1
Rack
VM #2
UD #2
VM #5
UD #5
Fault Domain 2
Rack
VM #3
UD #3
VM #6
UD #1

5. Region Pairs (Disaster Recovery Layer)

What is a Region Pair?

Each Azure region is paired with **another region in the same geography**.

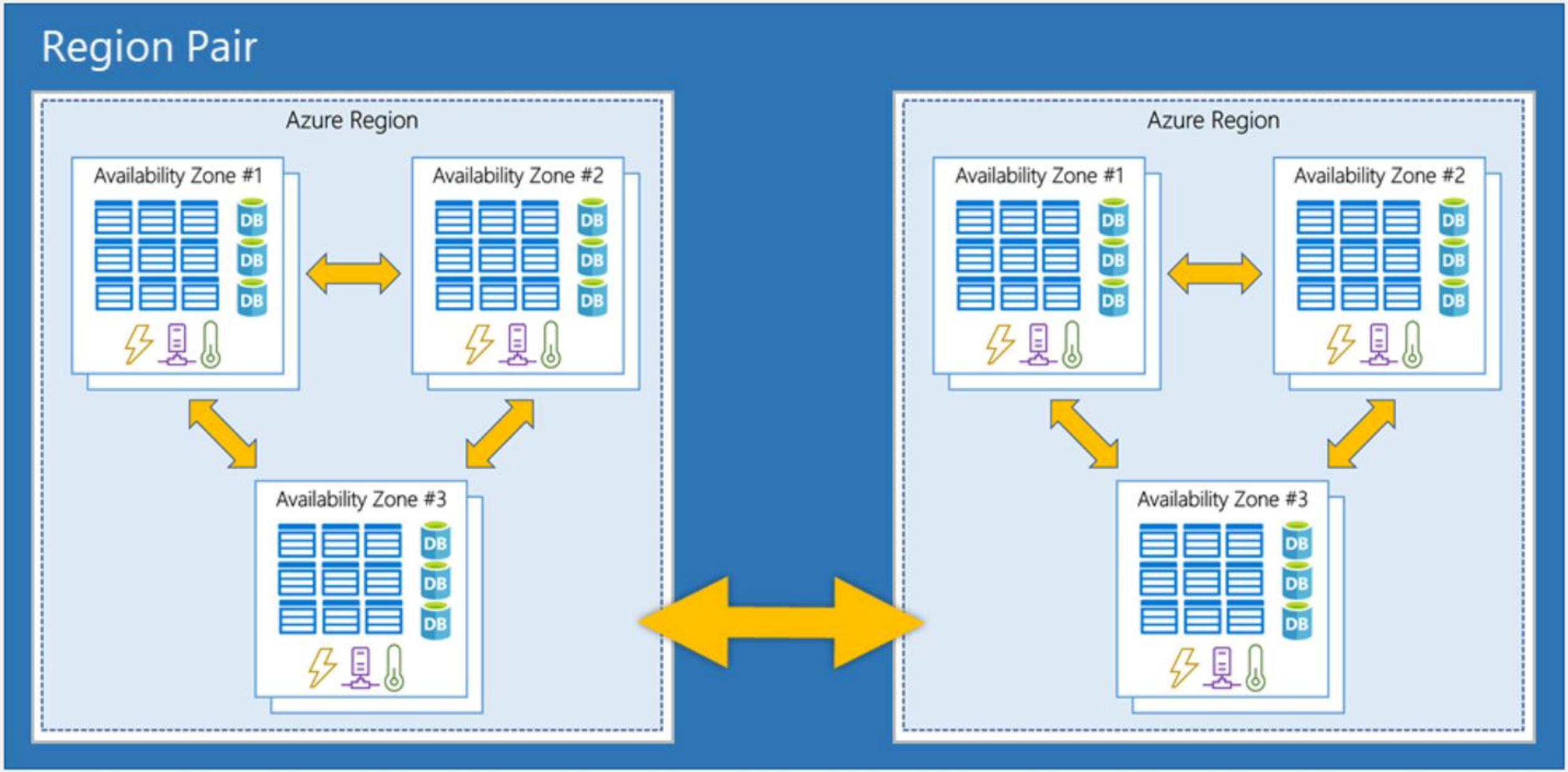
Characteristics

- Minimum **300 miles separation**
- Sequential updates (one region updated at a time)
- Automatic replication for some services

Examples

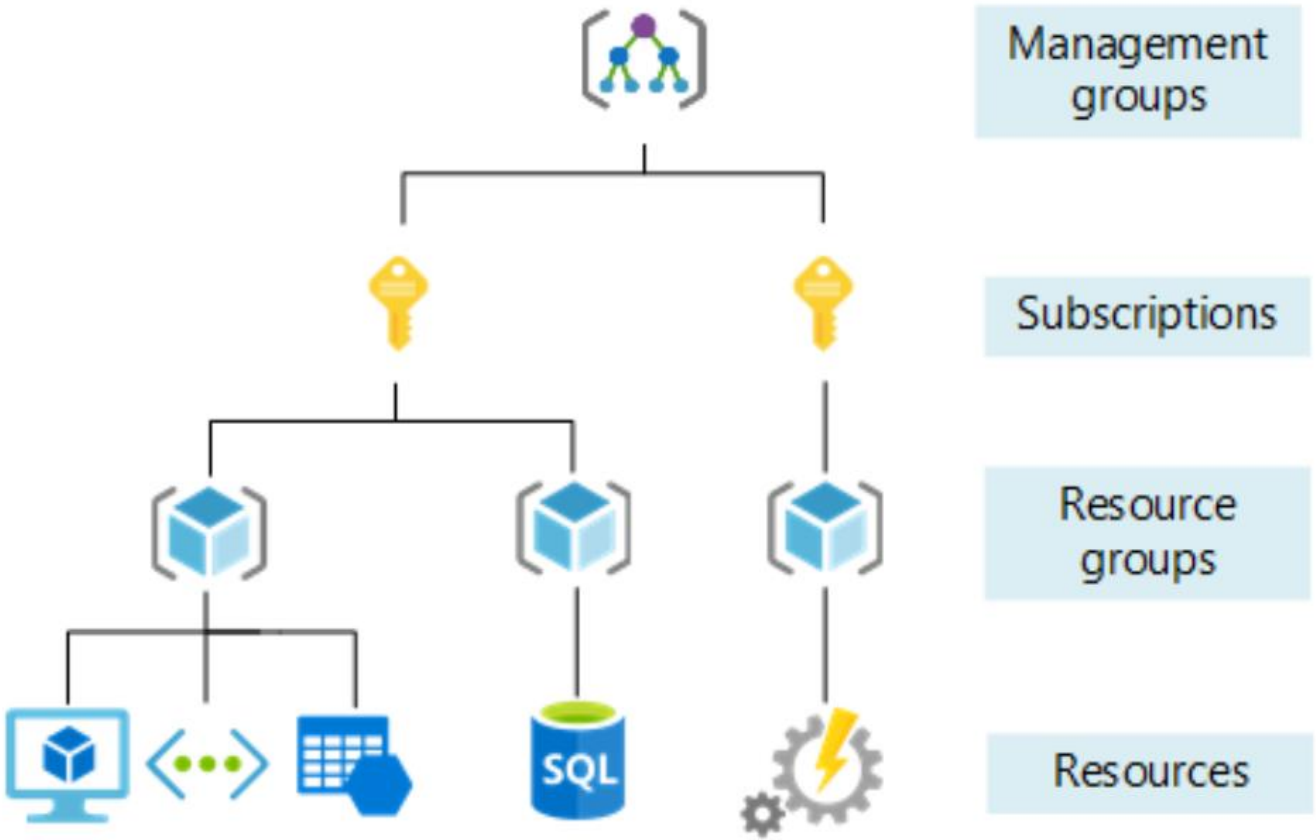
- West Europe ↔ North Europe
- East US ↔ West US

Geography


Region Pair
Azure Region
Availability Zone #1
DB
DB
DB
Availability Zone #2
DB
DB
DB
Availability Zone #3
DB
DB
DB
Azure Region
Availability Zone #1
DB
DB
DB
Availability Zone #2
DB
DB
DB
Availability Zone #3
DB
DB
DB

Azure management infrastructure

The management infrastructure includes Azure resources and resource groups, subscriptions, and accounts. Understanding the hierarchical organization will help you plan your projects and products within Azure.



Management Groups

- Top-most level
- Used to:
  - Apply policies
  - Manage multiple subscriptions
- Supports inheritance

📌 Large enterprises use this heavily

Subscriptions

What is a Subscription?

- **Billing boundary**
- **Access control boundary**

Key Facts

- One account can have multiple subscriptions
- Limits applied per subscription
- Used to separate:
  - Environments (Prod / Dev)
  - Teams
  - Projects

Azure resources and resource groups

A resource is the basic building block of Azure. Anything you create, provision, deploy, etc. is a resource. Virtual Machines (VMs), virtual networks, databases, cognitive services, etc. are all considered resources within Azure.

Resource groups are simply groupings of resources. When you create a resource, you’re required to place it into a resource group. While a resource group can contain many resources, a single resource can only be in one resource group at a time. Some resources may be moved between resource groups, but when you move a resource to a new group, it will no longer be associated with the former group. Additionally, resource groups can't be nested, meaning you can’t put resource group B inside of resource group A.

Azure Accounts

To create and use Azure services, you need an Azure subscription. When you're working with your own applications and business needs, you need to create an Azure account, and a subscription will be created for you. After you've created an Azure account, you're free to create additional subscriptions. For example, your company might use a single Azure account for your business and separate subscriptions for development, marketing, and sales departments. After you've created an Azure subscription, you can start creating Azure resources within each subscription.

Create an Azure account

Link :
https://azure.microsoft.com/en-us/pricing/purchase-options/azure-account

What is the Azure free account?

The Azure free account includes:

- Free access to popular Azure products for 12 months.
- A credit to use for the first 30 days.
- Access to more than 25 products that are always free.

The Azure free account is an excellent way for new users to get started and explore. To sign up, you need a phone number, a credit card, and a Microsoft or GitHub account. The credit card information is used for identity verification only. You won't be charged for any services until you upgrade to a paid subscription.

What is the Azure free student account?

The Azure free student account offer includes:

- Free access to certain Azure services for 12 months.
- A credit to use in the first 12 months.
- Free access to certain software developer tools.

The Azure free student account is an offer for students that gives $100 credit and free developer tools. Also, you can sign up without a credit card.